

Basic R

2026.05.10

Agenda

- Follow-up from day 1
 - Central terms
 - Readability and structure in scripts
 - Importing data
- Statistics in R
- Writing tables
- Pizza? 18.30?
- Generative AI as a tutor
- Final remarks

Summary of central terms

- Variable: A **column** in a dataset that represents a characteristic being measured (e.g. len)
- Value: A **element of data** within a variable for one observation (e.g. 4.2)
- Observation: A **row** in a dataset, representing one individual case or experimental unit (e.g. the row containing following values 4.2, VC, 0.5)
- Object: **named item** that stores data or represents something that you can use and manipulate in your code (e.g. `df <- ToothGrowth`).
- Function: **A piece of code** that performs a specific operation, usually taking inputs (arguments) and returning an output (e.g. `filter()`, `select()`, and `mutate()`)

len	supp	dose
4.2	VC	0.5
11.5	VC	0.5
7.3	VC	0.5
5.8	VC	0.5
6.4	VC	0.5
10.0	VC	0.5

Readable and structured coding - scripts

- Pipes (|>) at end of line (space before)
- New line per pipe
- Spaces on either side of mathematical operators apart from ^ (i.e. +, -, ==, <, ...), and around the assignment operator (<-)
- Use consistent naming (e.g. snake_case)

```
# Strive for
flights |>
  filter(!is.na(arr_delay), !is.na(tailnum)) |>
  count(dest)

# Avoid
flights|>filter(!is.na(arr_delay), !is.na(tailnum))|>count(dest)
```

Readable and structured coding - scripts

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- Spaces on either side of mathematical operators apart from ^ (i.e. +, -, ==, <, ...), and around the assignment operator (<-)
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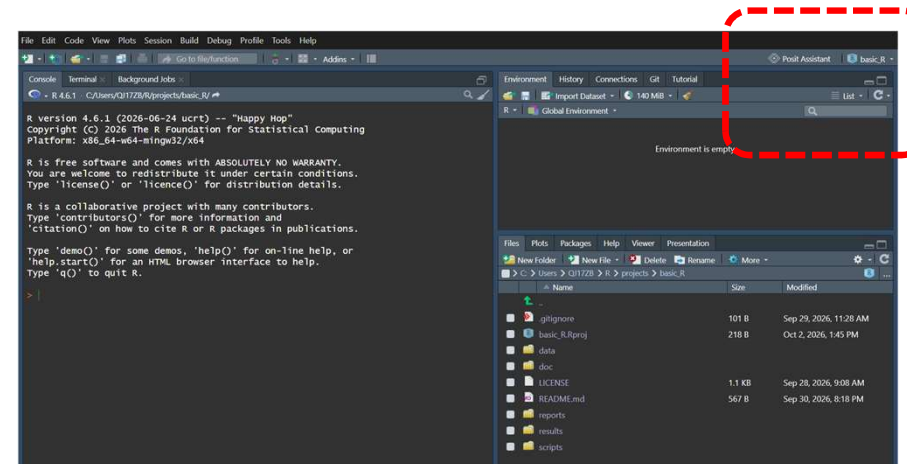
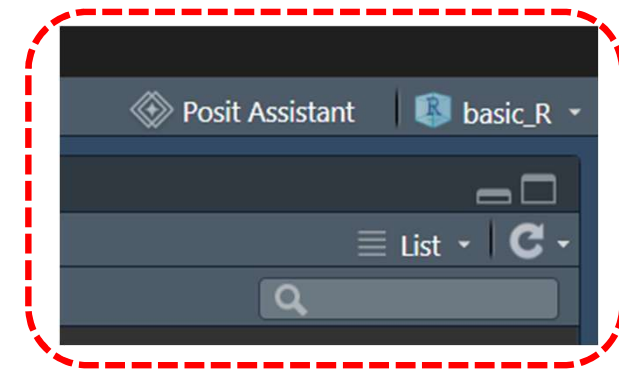
```
# Strive for
flights |>
  filter(!is.na(arr_delay), !is.na(tailnum)) |>
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# Avoid
flights|>filter(!is.na(arr_delay), !is.na(tailnum))|>count(dest)
```

What are the “issues” with second example?

Readable and structured coding - projects

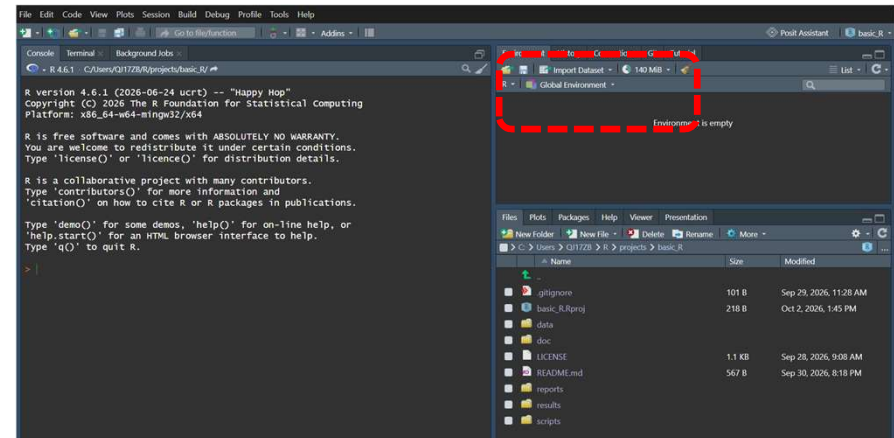
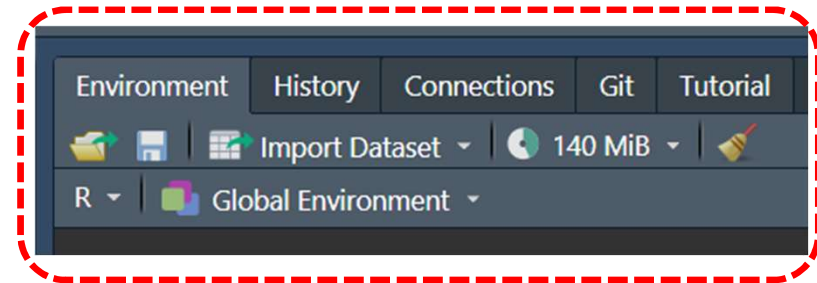
- Better organization: Keeps your scripts, data, results, and other files together in one dedicated folder.
- Easy file management: RStudio automatically works within the project's folder, **reducing problems with file paths**.
- Reproducibility: Makes it easier for others (and future self!) to run the analysis without changing file locations.



Importing data

“Import Dataset” functionality

- Good option to play around with separators, column names, e.g.
- Remember:
 - note down the selected options
 - Name the object according to scripts (e.g. df or gtex_v11_subjects)



Readable and structured coding

- Sectioning
- Document outline

As your scripts get longer, you can use **sectioning** comments to break up your file into manageable pieces:

```
# Load data -----  
  
# Plot data -----
```

RStudio provides a keyboard shortcut to create these headers (Cmd/Ctrl + Shift + R), and will display them in the code navigation drop-down at the bottom-left of the editor, as shown in [Figure 4.2](#).

Readable and structured coding

- Sectioning
- Document outline
- Hotkey for new section (Ctrl + Shift + R)
- Multiple # for subheaders

loading data
transforming data
significance testing with ...
create a contingency ta...
chi squared test

```
18 gtex_v11_subjects <- gtex_v11_subjects |>
19 mutate(sex_fm = if_else(sex == 1, "m", "f"))
20
21
22
23
24 # significance testing with chi squared test-----
25
26 # Null hypothesis (H0): death cause and sex are independent - the distribution of causes is the same
27 # Alternative hypothesis (H1): death cause and sex are associated - the distribution of causes differs
28
29 gtex_v11_subjects |>
30 count(sex_fm, OTMWDY) |>
31 pivot_wider(names_from = OTMWDY, values_from = n)
32
33
34
35 # create a contingency table
36
37
38
39 chisq.test(table(gtex_v11_subjects$sex_fm, gtex_v11_subjects$OTMWDY))
40
41
```

The screenshot shows the RStudio IDE with a script editor on the left, a console at the bottom, and a file explorer on the right. A red dashed box highlights a section of the script containing comments and code for significance testing. The console shows the output of the `chisq.test` function, which is a contingency table and a p-value.

	f	m
OTMWDY		
1	0	194
2	1	12
3	2	50
4	3	20
5	4	46

The console output shows the result of the `chisq.test` function, which is a contingency table and a p-value.

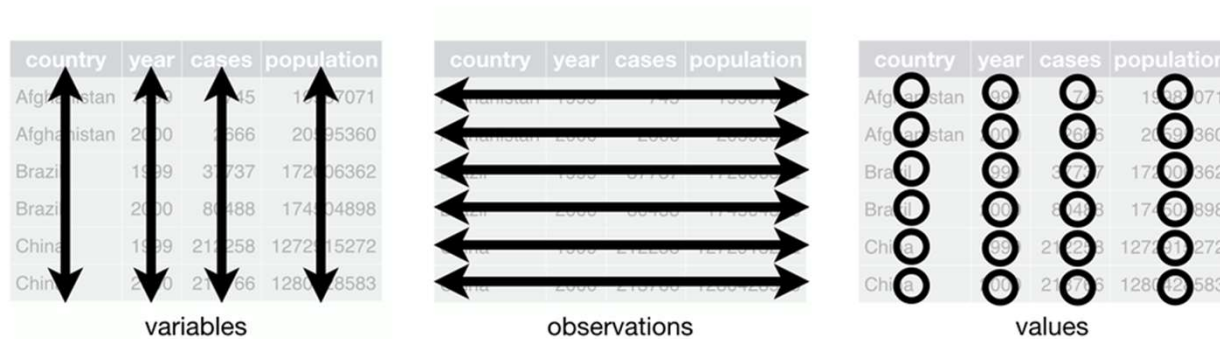
```
R> chisq.test(table(gtex_v11_subjects$sex_fm, gtex_v11_subjects$OTMWDY))
Chi-squared test for independence of sex_fm and OTMWDY
data:  table(gtex_v11_subjects$sex_fm, gtex_v11_subjects$OTMWDY)
X-squared = 1.144, df = 4, p-value = 0.977
```

Tidy data

There are three interrelated rules that make a dataset tidy:

1. Each variable is a column; each column is a variable.
2. Each observation is a row; each row is an observation.
3. Each value is a cell; each cell is a single value.

Figure 5.1 shows the rules visually.



Tidy data

- Purpose: Simple workflow for creating plots and perform calculations
- Recommendation: Report data in tidy format as part of data collection

country	year	cases	population
Afghanistan	1999	18145	19987071
Afghanistan	2000	18666	20095360
Brazil	1999	31737	17206362
Brazil	2000	34488	17404898
China	1999	212258	1272015272
China	2000	216766	1280625583

variables

country	year	cases	population
Afghanistan	1999	18145	19987071
Afghanistan	2000	18666	20095360
Brazil	1999	31737	17206362
Brazil	2000	34488	17404898
China	1999	212258	1272015272
China	2000	216766	1280625583

observations

country	year	cases	population
Afghanistan	1999	18145	19987071
Afghanistan	2000	18666	20095360
Brazil	1999	31737	17206362
Brazil	2000	34488	17404898
China	1999	212258	1272015272
China	2000	216766	1280625583

values

Tidy data

- How to calculate multiple summaries from multiple variables (e.g. mean and median from cases and population)?

```
> table1
# A tibble: 6 × 4
  country    year  cases population
  <chr>    <dbl> <dbl>    <dbl>
1 Afghanistan 1999     745  19987071
2 Afghanistan 2000    2666 20595360
3 Brazil      1999   37737 172006362
4 Brazil      2000   80488 174504898
5 China       1999  212258 1272915272
6 China       2000  213766 1280428583

> table3
# A tibble: 6 × 3
  country    year rate
  <chr>    <dbl> <chr>
1 Afghanistan 1999 745/19987071
2 Afghanistan 2000 2666/20595360
3 Brazil      1999 37737/172006362
4 Brazil      2000 80488/174504898
5 China       1999 212258/1272915272
6 China       2000 213766/1280428583
```

Statistics in R

<https://www.modernstatisticswithr.com/basicstatistics.html#the-three-cultures>

- Chapter 3

Commonly used functions from {stats} package:

- `t.test()` - performs one and two sample t-tests on vectors of data.
- `lm()` - fit linear model
- `cor()` - `cor` computes the correlation of `x` and `y` if these are vectors.
- `chisq.test()` performs chi-squared contingency table tests and goodness-of-fit tests.

GTEEx v11 meta data

https://github.com/SteffanChristiansen/basic_R/tree/main/scripts

- Save gtex_v11_day2.R in local scripts folder in basic_R project

Generative AI (GAI) for coding

Purpose:

- Enables simple transition from text to output with reduced experience for coding
- Enables one-to-one tutoring for learning settings

GAI for Socratic tutoring

“Socratic tutoring is a teaching method rooted in the philosophies of Socrates, emphasizing dialogue and questioning to nurture critical thinking.” – Fakour and Imani (2025)

Advantages:

- Use of GAI may improve multiple dimensions of learning including motivation, engagement, and learning performance (Badger et al, 2026)
- Immediate feedback on coding

GAI for Socratic tutoring

1. Open new window/chat in favorite GAI (e.g. Microsoft copilot)

2. Starting prompt:

You are my strict coding tutor. Do not write full solutions or final code for me. Instead, use the Socratic method, give me conceptual explanations, ask guiding questions, and provide small hints to help me solve problems on my own.

3. Prompt

I need to create a box plot in tidyverse using the tooth growth data.
How to approach problem?

Socratic method

```
</> R  
ggplot(data = _____, aes(x = _____, y = _____)) +  
  geom_____()
```

Non-Socratic method

```
</> R  
library(tidyverse)  
  
ggplot(ToothGrowth, aes(x = supp, y = len)) +  
  geom_boxplot()
```

Data – ChickWeight {datasets}

“The ChickWeight data frame has 578 rows and 4 columns from an experiment on the effect of diet on early growth of chicks.”

https://github.com/SteffanChristiansen/basic_R/tree/main/scripts

- Save chickweight.R in local scripts folder in basic_R project

Data – ChickWeight {datasets}

1. Solve task 1A and task 1B with the use of GAI (non-socratic – business as usual)
2. Solve task 2A-2F in prompt based on Socratic tutor

Socratic tutoring

1. Open new window/chat in favorite GAI (e.g. Microsoft copilot)
2. Starting prompt:
You are my strict coding tutor. Do not write full solutions or final code for me. Instead, use the Socratic method, give me conceptual explanations, ask guiding questions, and provide small hints to help me solve problems on my own.
3. Prompt
[I need to ... in tidyverse using the tooth growth data. How to approach problem]

Final remarks

- Nomenclature in R and tidyverse
- Basic functionalities in R studio
- Importing data
- Transforming data with tidyverse
- Creating figures
- Basic statistics
 - Descriptive statistics
 - Hypothesis testing
- Exporting data

Final remarks

Test of Socratic tutoring for basic R



Non-Socratic method

```
</> R  
  
library(tidyverse)  
  
ggplot(ToothGrowth, aes(x = supp, y = len)) +  
  geom_boxplot()
```

Socratic method

```
</> R  
  
ggplot(data = _____, aes(x = _____, y = _____)) +  
  geom_____()
```

Thank you for participating!